EXERCISE 11.5

1. Compute $P(X=k)$ for the binomial distribution, $B(n,p)$ where

   (i) $n=6,\ p=\frac{1}{3},\ k=3$ (ii) $n=10,\ p=\frac{1}{5},\ k=4$ (iii) $n=9,\ p=\frac{1}{2},\ k=7$

2. The probability that Mr.Q hits a target at any trial is $\frac{1}{4}$. Suppose he tries at the target 10 times. Find the probability that he hits the target (i) exactly 4 times (ii) at least one time.
3. Using binomial distribution find the mean and variance of X for the following experiments
   (i) A fair coin is tossed 100 times, and X denote the number of heads.
   (ii) A fair die is tossed 240 times, and X denote the number of times that four appeared.
4. The probability that a certain kind of component will survive a electrical test is $\frac{3}{4}$. Find the probability that exactly 3 of the 5 components tested survive.
5. A retailer purchases a certain kind of electronic device from a manufacturer.
   The manufacturer indicates that the defective rate of the device is 5%.
   The inspector of the retailer randomly picks 10 items from a shipment. What is the probability that there will be (i) at least one defective item (ii) exactly two defective items?
6. If the probability that a fluorescent light has a useful life of at least 600 hours is 0.9, find the probabilities that among 12 such lights
   (i) exactly 10 will have a useful life of at least 600 hours;
   (ii) at least 11 will have a useful life of at least 600 hours;
   (iii) at least 2 will *not* have a useful life of at least 600 hours.
7. The mean and standard deviation of a binomial variate X are respectively 6 and 2.
   Find (i) the probability mass function (ii) $P(X=3)$ (iii) $P(X\geq 2)$.
8. If $X\sim B(n,p)$ such that $4P(X=4)=P(X=2)$ and $n=6$. Find the distribution, mean and standard deviation of X.
9. In a binomial distribution consisting of 5 independent trials, the probability of 1 and 2 successes are 0.4096 and 0.2048 respectively. Find the mean and variance of the random variable.

EXERCISE 11.6

Choose the Correct or the most suitable answer from the given four alternatives :

1. Let X be random variable with probability density function

$$f(x)=\begin{cases}\frac{2}{x^3} & x\geq 1\\ 0 & x<1\end{cases}$$

WLIQ7

Which of the following statement is correct?

(1) both mean and variance exist (2) mean exists but variance does not exist

(3) both mean and variance do not exist (4) variance exists but Mean does not exist.

12th_Maths_Vol 2_EM_CH 11_Probability Distributions_29-05-20.indd 218 16-12-2022 16:52:45